Prime-field leads 20261031: reruns of L2a, L2b and C2 (msolve leg)

Date. 2026-11-01. **Plan.** `RERUN_PLAN.md`, frozen before any data, with amendments A1 to A5 and deviation D1. **Execution.** Four task cards compiled by the experiment-director tier (`cards/`), each run with `tools/task_card.py run` from a checkout detached at the pinned commit; receipts in `receipts/`. The director's assessment (`ASSESSMENT.md`) classifies all four as executed as specified: every command exit 0 inside its timeout, every receipt digest and declared output present, branch copies byte-identical to the run outputs. `RERUN_PLAN.md` gained deviation D1 after the runs, so its digest on the branch no longer matches the one the cards bind; the runs used the bound version. **Host.** One cloud container, 4 vCPU x86-64, 15 GiB, no other workload. All metrics are counted (field multiplications, F4 step degrees, matrix sizes, round counts); wall time appears only as the kill caps. No speedup, exponent change or scoreboard change is claimed.

Outcome

Lead	Before this rerun	Now	Decision (frozen rule)
L2a	partial, s = 6..11	complete, s = 6..12, every attempt inside the guard	does not reopen; G closes (e = 1.97 [1.95, 2.00]); S at 2.19 [2.16, 2.22], above 2
L2b	1 ladder cell, no decision	70 cells, 68 completed	null_confirmed: trajectory ratio 4.00 (real) and 4.00 (control)
C2, msolve leg on M	5 of 16 cells completed, prefix comparison	16 of 16 completed (plus 16 extra cells)	not separated at completion and on the 95-round prefix: C2 stays null_confirmed

Instrument parity (stage 0)

The original host ran Homebrew msolve of unrecorded version on macOS arm64. This host runs msolve 0.6.5 (Ubuntu package). Building a newer msolve from source was refused by the session's permission policy.

Check	Cells	Result
First P1 attempt, default kernel (`attempt1_msolve065_l2/`)	32 C2 S cells	every cell segfaulted in its first F4 round (characteristic 105037 > 2^16)
First P2 attempt, default kernel	3 L2b probes (p < 2^16)	identical to the original in every column
P1 with `-l 1` (amendment A5)	32 C2 S cells	rounds, maximal degree, largest matrix and summed cells identical to the original; all 32 reduced bases identical
P2 with `-l 1`	3 L2b probes	identical in every column

C2 M reproduction (stage 2)	16 original M records	identical over every round the original recorded (5 completed, 11 cut at 420 s)
L2b ladder reproduction (stage 1)	1 original ladder cell (p = 65521, control, seeds 1/1)	identical
L2a reproduction (stage 3)	45 committed per-(arm, class, s) rows, s = 6..11	identical (attempt counts, mean work, mean pairs, multiplications per pair)

So the instrument here is the instrument of the original run, to every recorded digit.

L2a: slope e of counted work per attempt at m = 3 (s = 6..12)

Harness `../L2a_slope_e_m3/src/l2a.rs`, seeds unchanged; 3 curves × 32 targets per s; arms G (group meet-in-the-middle), S (algebraic S_4 root finding), T (triple scan, s ≤ 8). Fit `../L2a_slope_e_m3/src/stats.rs`: OLS of $\log_2$ mean work on s, hierarchical bootstrap (curves, then attempts), 4000 replicates. Files: `L2a/attempts.csv`, `L2a/results/fit.csv`, `L2a/results/per_s_means.csv`, graph `L2a/results/L2a_work_vs_s.svg`.

s	$\log_2$ p	G mean work	S mean work	S multiplications per pair
6	18.66	$1.02 \cdot 10^4$	$8.56 \cdot 10^5$	2143
7	21.54	$4.77 \cdot 10^4$	$6.09 \cdot 10^6$	2832
8	24.86	$1.71 \cdot 10^5$	$2.51 \cdot 10^7$	3118
9	27.19	$6.53 \cdot 10^5$	$1.05 \cdot 10^8$	3341
10	30.42	$2.60 \cdot 10^6$	$4.97 \cdot 10^8$	3992
11	33.38	$1.04 \cdot 10^7$	$2.04 \cdot 10^9$	4037
12	36.68	$4.03 \cdot 10^7$	$9.29 \cdot 10^9$	4804

Arm	Class	Points	e	95 % interval
G	all	7	1.97	[1.95, 2.00]
G	satisfiable	7	1.98	[1.82, 2.22]
G	unsatisfiable	7	1.97	[1.94, 2.00]
S	all	7	2.19	[2.16, 2.22]
S	satisfiable	7	2.18	[2.02, 2.43]
S	unsatisfiable	7	2.19	[2.15, 2.21]
T	all	3	3.21	[2.99, 3.40]

Attempts: 672 G, 672 S, 288 T, all `OK`; no target was cut by the 20-minute guard (the longest cell, S at s = 12, curve 1, took 621 s). With s = 12 added, the intervals moved by at most 0.02 from the s = 6..11 values reported earlier (G 1.98 [1.94, 2.01], S 2.21 [2.16, 2.24]).

Decision. Reopen needs an interval entirely below 0.5: not met on any arm or class. Close needs an interval containing 2: met for G (all and unsatisfiable; satisfiable [1.82, 2.22] too). S lies above 2 and is recorded as observed. Null prediction e ≥ 1: met everywhere.

L2b: d_reg trajectory of the constrained prime-field Semaev system

(catalogue M008) Ladder exactly as `../L2b_dreg_trajectory/run_ladder.sh` lists it (amendment A2: long cells 2 in parallel, short cells 4 in parallel, cap 1500 s and 6 GiB per run). Files: `L2b/results/cells.csv` (one row per cell; header in `../L2b_dreg_trajectory/results_header.txt`), `L2b/results/summary.csv`, `L2b/results/instances.jsonl`, gzipped msolve logs in `L2b/work/`; graph `L2b/dreg_vs_log2p.svg`.

p	$\log_2 p$	N	arm	runs	completed	median max degree	median widest matrix	median total rows	rounds (range)
1021	10.0	10	real	8	8	18	867 × 872	10191	101–104
1021	10.0	10	control	8	8	18	725 × 764	8078	23–37
4093	12.0	16	real	8	8	24	2059 × 2033	39534	252–274
4093	12.0	16	control	8	8	24	1844 × 1890	24235	35–55
16381	14.0	25	real	8	8	33	3413 × 5267	146060	557–588
16381	14.0	25	control	8	8	33	4480 × 4587	78402	53–84
65521	16.0	40	real	8	8	48	18227 × 18181	1449769	1584–1630
65521	16.0	40	control	8	8	48	11744 × 11790	287213	83–126
262139	18.0	64	real	2	2	72	35553 × 47832	$9.95 \cdot 10^6$	4068–4135
262139	18.0	64	control	2	2	72	7223 × 30330	$4.60 \cdot 10^5$	131
1048573	20.0	102	real	1	0 (cap 1500 s)	110 reached	18822 × 81222 reached	$2.64 \cdot 10^7$ reached	2211 at kill
1048573	20.0	102	control	1	0 (cap 1500 s)	110 reached	15794 × 77355 reached	$1.24 \cdot 10^6$ reached	119 at kill

Decision (PREREG.md). Largest p where both arms completed every run: 262139. Trajectory ratio (median maximal degree there over the same at p = 1021): real 72 / 18 = 4.00, control 72 / 18 = 4.00. The reopen clause (real < 1.5 while control ≥ 2) is not met; the two ratios agree within 25 % (they are equal), so the decision is **null_confirmed**. In every completed cell of either arm the maximal step degree is N + 8, which `PREREG.md` said to expect from the membership polynomials.

Observed, not decided. The real arm does more F4 work than the control at every rung, and the gap grows with p: the real-to-control ratio of median total rows is 1.26, 1.63, 1.86, 5.05 and 21.6 from

p = 1021 to p = 262139, and the real arm takes about 3 to 31 times as many rounds (median 104 against 37 at p = 1021, about 4100 against 131 at p = 262139). The interval-constrained S_4 system is harder for F4 than a random system of the same shape, not easier. The two p = 1048573 cells were stopped by the time cap; no cell reached the 6 GiB memory cap (peak memory was not recorded).

C2: msolve leg on the M instance (crater against floor)

The 32 committed M inputs (`../C2_volcano/M/msolve/*.ms`, hashes in `C2_M/inputs.sha256`), msolve `-l 1`, 4 threads, cap 3600 s (amendment A3). Files: `C2_M/msolve_summary.csv`, `C2_M/msolve_trajectories.csv`, `C2_M/msolve/*.out`; graph `C2_M/graph_msolve_trajectory.svg` (script `plot_c2m.py`, recorded values only). The frozen rule uses the 16 cells of models 0 to 3; the 16 cells of models 4 to 7 ran because the driver copied every committed input (deviation D1) and enter no decision.

Cells	Side	Completed	Max step degree	Rounds	Summed matrix cells	Widest matrix (columns)
models 0–3 (decision)	crater	8 / 8	55 in every cell	2037–2089	$2.06{\cdot}10^{10}$ – $2.78{\cdot}10^{10}$	17380 – 28506
models 0–3 (decision)	floor	8 / 8	55 in every cell	2037–2089	$2.06{\cdot}10^{10}$ – $2.78{\cdot}10^{10}$	17380 – 28506
models 4–7 (extra)	crater	8 / 8	55	2037–2089	$2.06{\cdot}10^{10}$ – $2.78{\cdot}10^{10}$	17380 – 28506
models 4–7 (extra)	floor	8 / 8	55	2037–2089	$2.06{\cdot}10^{10}$ – $2.78{\cdot}10^{10}$	17380 – 28506

Decision (frozen rule). (a) Common prefix, as frozen: over the first 95 rounds the summed matrix cells are 3079461630 in 15 of the 16 decision cells and 3079459926 in crater_3_t1, maximal degree 55 in all, as originally reported. (b) At completion: every crater cell and every floor cell has maximal step degree 55, and the round counts and matrix profiles span the same ranges on both sides. No crater cell differs from every floor cell by one degree, so the msolve leg is **not separated** under both readings. With the counted-work leg unchanged (crater-to-floor ratio 0.9998 on M), C2 stays **null_confirmed**, now with all 16 msolve cells completed instead of 5.

Graphs

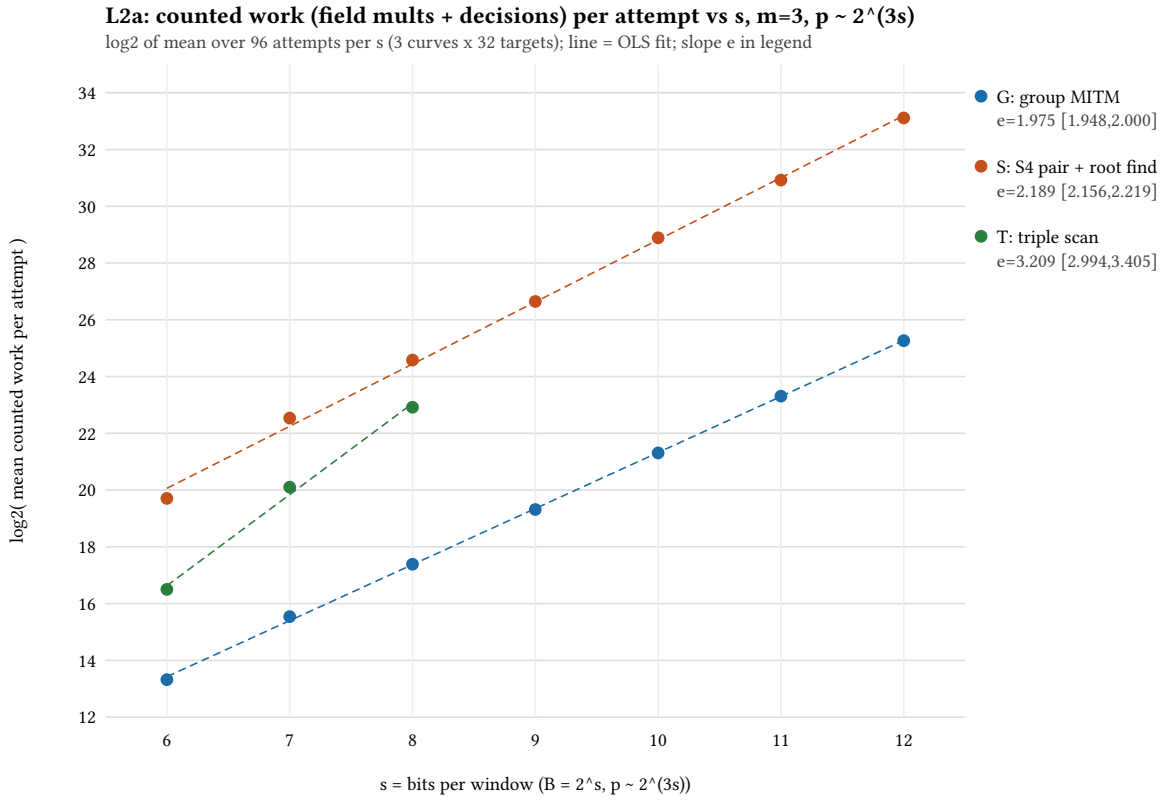


Figure 1: L2a: $\log_2$ mean counted work per attempt against s, with fitted slopes

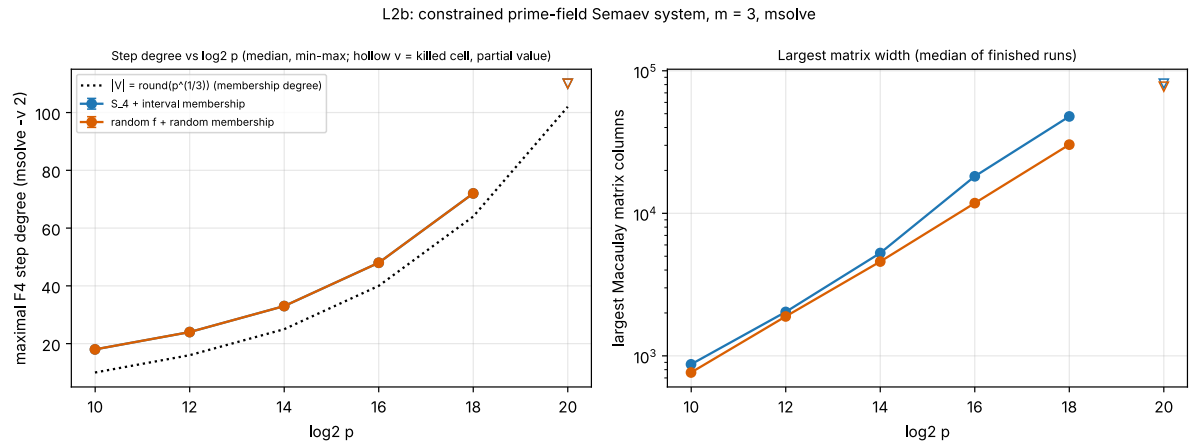


Figure 2: L2b: maximal F4 step degree and widest matrix against $\log_2$ p, real arm and control

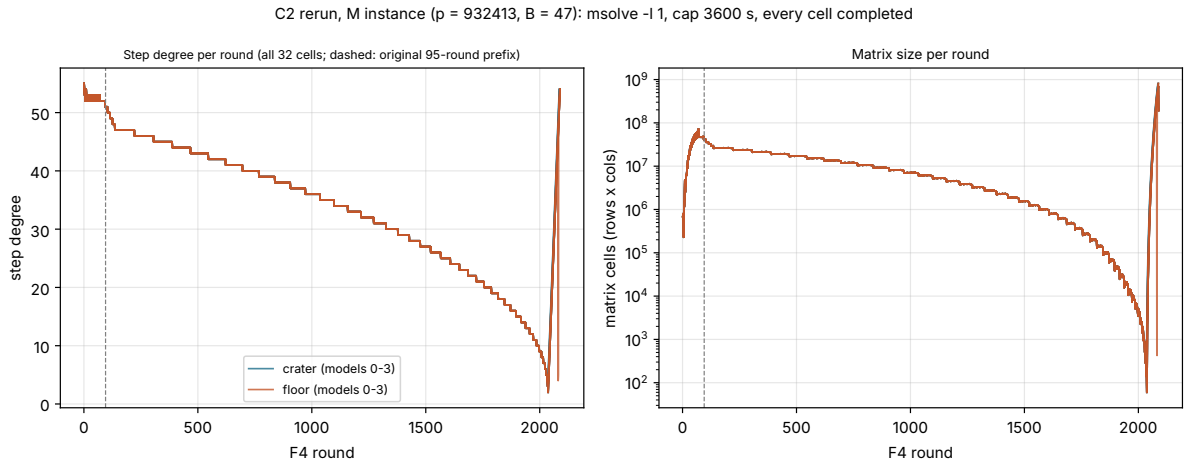


Figure 3: C2 M: step degree and matrix size per F4 round, crater and floor

Deviations and limits

- Route: not Iso Lab (`RERUN_PLAN.md`, "Host and route"). The resume checkpoint went to the branch, not S3; launches carried `CHECKPOINT_SKIP` with that reason.
- Execution tier: the main session ran the cards through the task-card tool instead of relaying them to the mechanical executor, whose background launch the checkpoint hook would refuse here. The tool, not a model, executed every command.
- msolve kernel `-l 1` throughout (A5); parity above shows it reproduces the original records on 35 + 16 + 1 cells.
- L2b memory cap 6 GiB instead of 16 GiB (A2); no cell reached it.
- C2 M ran 32 cells instead of 16 (D1); the extra 16 enter no decision.
- L2a: one prime per s and three curves per prime, as frozen; the SAT/SMT decomposers named in the lead text remain unmeasured.

Reading

The three open cells of the round are closed in the direction the round predicted. At m = 3 the counted cost per decomposition attempt grows as B^2 for the group meet-in-the-middle and slightly faster for the algebraic S_4 arm, four times the exponent that would beat rho. The constrained Semaev system's step degree tracks the membership degree exactly as a random system's does, and F4 does more work on it than on the random control. On one volcano the crater and floor curves give identical F4 profiles. None of this moves the index-calculus baseline. Distance remaining to rho at m = 3: the measured slope is $e \approx 2.0$ against the $e < 0.5$ that §2 of the leads note requires, a gap of about 1.5 in e, unchanged by s = 12.